

Probability Worked Solutions & the CLT

Solutions to the conditional probability and Bayes’ theorem problems, plus the Central Limit Theorem written in terms of the normal distribution

1. Conditional Probability — the Spam / “Free” Example

Given

- 100 emails total
- 20 are spam $\rightarrow P(\text{spam}) = 20/100 = 0.20$
- 15 of those 20 spam emails contain the word “free” $\rightarrow P(\text{free} \mid \text{spam}) = 15/20 = 0.75$

Question: given that an email contains “free,” what fraction are spam? That is, find **$P(\text{spam} \mid \text{free})$** .

$$P(\text{spam} \mid \text{free}) = P(\text{spam} \cap \text{free}) / P(\text{free})$$

Why this can’t be solved from the numbers given alone: we know the numerator, $P(\text{spam} \cap \text{free}) = 15/100 = 0.15$ (15 emails are both spam *and* contain “free”). But the denominator, $P(\text{free})$, is the *total* fraction of emails containing “free” — spam **and** non-spam combined. The original example only describes the spam subset; it never states how many of the 80 non-spam emails also contain “free.” Without that, $P(\text{free})$ is unknown. The example illustrates the *concept* of narrowing the sample space, rather than supplying enough data to compute a number.

Making it solvable

Suppose we are additionally told: *5 of the 80 non-spam emails also contain “free.”*

Step 1 — total probability of containing “free” (from both spam and non-spam emails):

$$P(\text{free}) = (15 + 5) / 100 = 0.20$$

Step 2 — apply the conditional probability formula:

$$P(\text{spam} \mid \text{free}) = 0.15 / 0.20 = 0.75$$

Answer: with that added assumption, 75% of emails containing “free” would be spam.

2. Bayes’ Theorem — the Medical Test Example

Given

- $P(\text{disease}) = 1\% \rightarrow P(D) = 0.01$, so $P(\text{not } D) = 0.99$
- The test is “99% accurate,” read as: $P(\text{positive} \mid D) = 0.99$ (true positive rate) and $P(\text{positive} \mid \text{not } D) = 0.01$ (false positive rate, i.e. 99% specificity too)

Question: given a positive test result, what is $P(D \mid \text{positive})$ — the probability the person actually has the disease?

$$P(D \mid +) = [P(+ \mid D) \cdot P(D)] / P(+)$$

Step 1 — total probability of testing positive, from both paths (true positives + false positives):

$$P(+) = P(+ \mid D)P(D) + P(+ \mid \text{not } D)P(\text{not } D)$$
$$P(+) = (0.99)(0.01) + (0.01)(0.99) = 0.0099 + 0.0099 = 0.0198$$

Step 2 — plug into Bayes’ theorem:

$$P(D \mid +) = 0.0099 / 0.0198 = 0.50$$

Answer: $P(\text{disease} \mid \text{positive test}) = 50\%$

Why this is counter-intuitive: even with a “99% accurate” test, a positive result only means a coin-flip’s chance of actually having

the disease. Since only 1% of people have the disease, the pool of healthy people is far larger — so even a 1% false-positive rate among them produces roughly as many false positives (0.0099) as true positives (0.0099). The rare prior, $P(D) = 0.01$, is what drags the answer down from 99% to 50%.

3. The Central Limit Theorem, Written in Terms of the Normal Distribution

The **Central Limit Theorem (CLT)** is the formal reason the normal distribution shows up so often: it states that a sum (or average) of many independent random variables tends toward a normal distribution, *regardless of the original variables' own distribution*.

For i.i.d. random variables $X_1, X_2, \dots, X_n$ with mean μ and finite variance σ^2 :

Sample mean form

$$\bar{X}_n = (1/n) \sum X_i \rightarrow N(\mu, \sigma^2/n) \text{ as } n \rightarrow \infty$$

For large n , the sample mean is *approximately* normally distributed with mean μ and variance σ^2/n — so its spread shrinks as n grows.

Standardized form

The more formal statement of convergence:

$$(\bar{X}_n - \mu) / (\sigma/\sqrt{n}) \rightarrow N(0, 1)$$

The standardized sample mean converges in distribution to the **standard normal** exactly.

Sum form

Equivalent to the sample mean form, just unnormalized:

$$S_n = \sum X_i \approx N(n\mu, n\sigma^2) \text{ for large } n$$

This connects directly to the normal probability density function:

$$f(x) = [1 / (\sigma\sqrt{2\pi})] \cdot e^{-(x-\mu)^2 / (2\sigma^2)}$$

The CLT is *why* this bell-curve formula — with parameters μ and σ^2/n — is the right approximation to use for a sample mean, even when the underlying X_i come from a completely different (e.g. skewed, discrete, or bounded) distribution.